

Platform Technologies

Mobile Platforms

(Android / IOS Basics)

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Year and Section:

IT - 41

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Basic Information:

I have created a calculator that can only solve 2 numbers by using addition, subtraction, multiplication, and division. Unlike in our current calculator, it can solve multiple numbers, but my work can only solve 2 numbers, if you want to proceed with a new number, you need to input another number to solve a new problem. There's also a feature where you can click the dark mode button that can turn the background into black and the buttons to orange.

Screenshot/s of the Code:

activity_main.xml | MainActivity.kt | colors.xml

OFF

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical">

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:layout_margin="5dp"
        android:gravity="end"
        android:orientation="horizontal">

        <TextView
            android:layout_width="0dp"
            android:layout_height="wrap_content"
            android:layout_weight="1"
            android:id="@+id/JohnTv"
            android:text="Johns Calculator"
            android:textSize="30sp"
            android:fontFamily="serif"/>


        <Button
            android:id="@+id/dark_mode"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Dark Mode"
            android:backgroundTint="@color/black"/>

    </LinearLayout>

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/et_a"
        android:layout_margin="20dp"
```

activity_main.xml | MainActivity.kt | colors.xml

OFF

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    </LinearLayout>

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/et_a"
        android:layout_margin="20dp"
        android:hint="Enter a number"
        android:textSize="30sp"/>

    <EditText
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:id="@+id/et_b"
        android:layout_margin="20dp"
        android:hint="Enter a number"
        android:textSize="30sp"/>

    <LinearLayout
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:orientation="horizontal"
        android:gravity="center">

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/btn_add"
            android:layout_marginRight="5dp"
            android:text="+"
            android:backgroundTint="@color/black" />

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/btn_sub"
            android:layout_marginRight="5dp"
```

activity_main.xml | MainActivity.kt | colors.xml

OFF

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    <LinearLayout

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/btn_sub"
            android:layout_marginRight="5dp"
            android:text="-"
            android:backgroundTint="@color/black"/>

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/btn_multiply"
            android:layout_marginRight="5dp"
            android:text="*"
            android:backgroundTint="@color/black"/>

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/btn_division"
            android:text="/"
            android:backgroundTint="@color/black"/>

    </LinearLayout>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Result:"
        android:textSize="24sp"
        android:gravity="center"
        android:id="@+id/result_tv"
        android:layout_margin="20dp"
        android:background="@color/black"
        android:textColor="@color/white"/>
```

activity_main.xml | MainActivity.kt | colors.xml

OFF

```
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    <LinearLayout
        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/btn_multiply"
            android:layout_marginRight="5dp"
            android:text="*"
            android:backgroundTint="@color/black"/>

        <Button
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:id="@+id/btn_division"
            android:text="/"
            android:backgroundTint="@color/black"/>

    </LinearLayout>

    <TextView
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Result:"
        android:textSize="24sp"
        android:gravity="center"
        android:id="@+id/result_tv"
        android:layout_margin="20dp"
        android:background="@color/black"
        android:textColor="@color/white"/>



</LinearLayout>
```

activity_main.xml | MainActivity.kt | colors.xml

```kotlin
package com.example.mykotlinapp

import android.os.Bundle
import android.view.View
import android.widget.Button
import android.widget.EditText
import android.widget.TextView
import androidx.activity.enableEdgeToEdge
import androidx.appcompat.app.AppCompatActivity
import androidx.core.view.ViewCompat
import androidx.core.view.WindowInsetsCompat
import androidx.core.view.WindowInsetsControllerCompat


1 Usage
class MainActivity : AppCompatActivity(), View.OnClickListener {

    4 Usages
    lateinit var btnAdd: Button
    4 Usages
    lateinit var btnSub: Button
    4 Usages
    lateinit var btnMultiply: Button
    4 Usages
    lateinit var btnDivision: Button
    4 Usages
    lateinit var darkModeBtn: Button
    6 Usages
    lateinit var etA: EditText
    6 Usages
    lateinit var etB: EditText
    4 Usages
    lateinit var resultTv: TextView
    3 Usages
    lateinit var JohnTv: TextView

    3 Usages
    var isDarkMode = false
```

activity_main.xml | MainActivity.kt | colors.xml

```
15      class MainActivity : AppCompatActivity(), View.OnClickListener {
27          var isDarkMode = false
28
29          override fun onCreate(savedInstanceState: Bundle?) {
30              super.onCreate(savedInstanceState)
31              enableEdgeToEdge()
32              setContentView(R.layout.activity_main)
33
34              ViewCompat.setOnApplyWindowInsetsListener( view = findViewById(R.id.main)) { v, insets ->
35                  val systemBars = insets.getInsets( typeMask = WindowInsetsCompat.Type.systemBars())
36                  v.setPadding(
37                      systemBars.left,
38                      systemBars.top,
39                      systemBars.right,
40                      systemBars.bottom
41                  )
42                  insets
43              }
44
45              btnAdd = findViewById(R.id.btn_add)
46              btnSub = findViewById(R.id.btn_sub)
47              btnMultiply = findViewById(R.id.btn_multiply)
48              btnDivision = findViewById(R.id.btn_division)
49
50              darkModeBtn = findViewById(R.id.dark_mode)
51
52
53              etA = findViewById(R.id.et_a)
54              etB = findViewById(R.id.et_b)
55              resultTv = findViewById(R.id.result_tv)
56              JohnTv = findViewById(R.id.JohnTv)
57
58              btnAdd.setOnClickListener(this)
59              btnSub.setOnClickListener(this)
60              btnMultiply.setOnClickListener(this)
61              btnDivision.setOnClickListener(this)
62
63              darkModeBtn.setOnClickListener {
```

activity_main.xml | MainActivity.kt | colors.xml

```
15      class MainActivity : AppCompatActivity(), View.OnClickListener {
29          override fun onCreate(savedInstanceState: Bundle?) {
63              darkModeBtn.setOnClickListener {
65
66                  if (isDarkMode) {
67
68                      findViewById<View>(R.id.main).setBackgroundColor(getColor( id = R.color.black))
69
70                      WindowInsetsControllerCompat(window, window.decorView)
71                          .isAppearanceLightStatusBars = false
72
73                      btnAdd.backgroundTintList = getColorStateList( id = R.color.orange)
74                      btnSub.backgroundTintList = getColorStateList( id = R.color.orange)
75                      btnMultiply.backgroundTintList = getColorStateList( id = R.color.orange)
76                      btnDivision.backgroundTintList = getColorStateList( id = R.color.orange)
77                      resultTv.backgroundTintList = getColorStateList( id = R.color.orange)
78                      darkModeBtn.backgroundTintList = getColorStateList( id = R.color.orange)
79
80                      etA.setTextColor(getColor( id = R.color.white))
81                      etB.setTextColor(getColor( id = R.color.white))
82                      JohnTv.setTextColor(getColor( id = R.color.white))
83
84                      etA.setHintTextColor(getColor( id = R.color.white))
85                      etB.setHintTextColor(getColor( id = R.color.white))
86
87
88
89                      window.statusBarColor = getColor( id = R.color.black)
90
91                  } else {
92                      findViewById<View>(R.id.main).setBackgroundColor(getColor( id = R.color.white))
93
94                      WindowInsetsControllerCompat(window, window.decorView)
95                          .isAppearanceLightStatusBars = true
96
97                      btnAdd.backgroundTintList = getColorStateList( id = R.color.black)
98                      btnSub.backgroundTintList = getColorStateList( id = R.color.black)
99                      btnMultiply.backgroundTintList = getColorStateList( id = R.color.black)
```

activity_main.xml | MainActivity.kt | colors.xml

```
15      class MainActivity : AppCompatActivity(), View.OnClickListener {
29          override fun onCreate(savedInstanceState: Bundle?) {
63              darkModeBtn.setOnClickListener {
98                  btnDivision.backgroundTintList = getColorStateList( id = R.color.black)
99                  resultTv.backgroundTintList = getColorStateList( id = R.color.black)
100                 darkModeBtn.backgroundTintList = getColorStateList( id = R.color.black)
101
102                 etA.setTextColor(getColor( id = R.color.black))
103                 etB.setTextColor(getColor( id = R.color.black))
104                 JohnTv.setTextColor(getColor( id = R.color.black))
105
106                 etA.setHintTextColor(getColor( id = R.color.gray))
107                 etB.setHintTextColor(getColor( id = R.color.gray))
108
109
110
111             }
112         }
113     }
114
115     override fun onClick(v: View?) {
116
117         var a = etA.text.toString().toDouble()
118         var b = etB.text.toString().toDouble()
119         var result = 0.0
120
121         when (v?.id) {
122             R.id.btn_add -> {
123                 result = a + b
124             }
125
126             R.id.btn_sub -> {
127                 result = a - b
128             }
129
130             R.id.btn_multiply -> {
131                 result = a * b
132             }
```

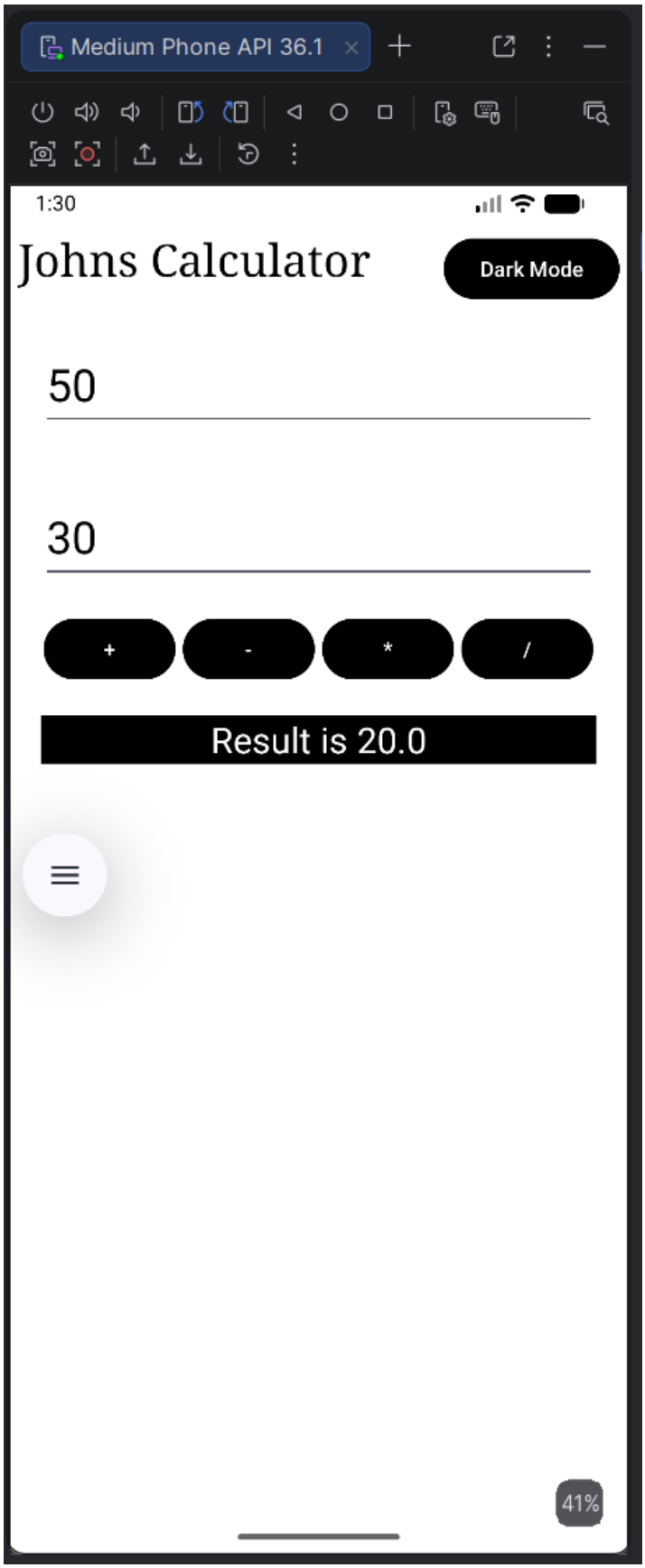
activity_main.xml | MainActivity.kt | colors.xml

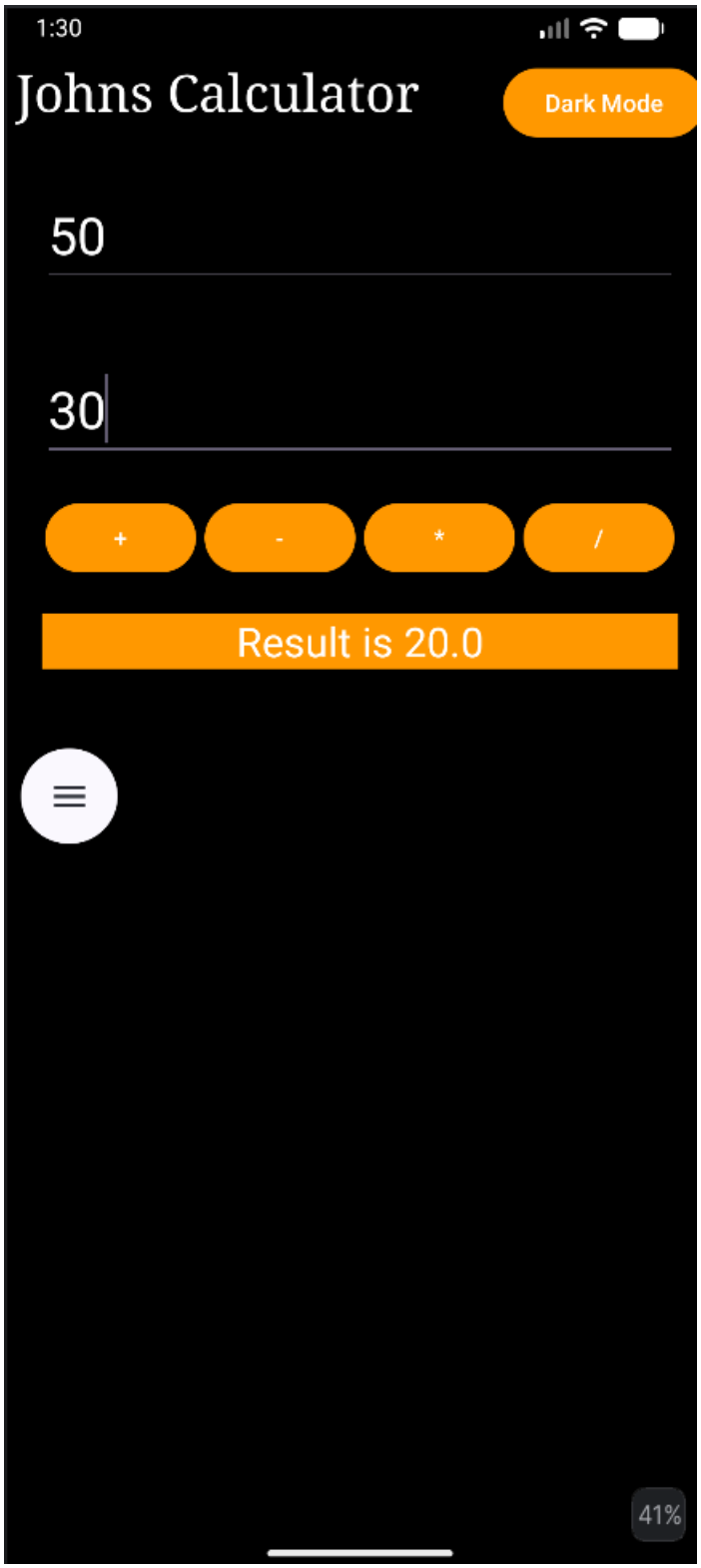
```
15      class MainActivity : AppCompatActivity(), View.OnClickListener {
29          override fun onCreate(savedInstanceState: Bundle?) {
63              darkModeBtn.setOnClickListener {
112             }
113         }
114
115         override fun onClick(v: View?) {
116
117             var a = etA.text.toString().toDouble()
118             var b = etB.text.toString().toDouble()
119             var result = 0.0
120
121             when (v?.id) {
122                 R.id.btn_add -> {
123                     result = a + b
124                 }
125
126                 R.id.btn_sub -> {
127                     result = a - b
128                 }
129
130                 R.id.btn_multiply -> {
131                     result = a * b
132                 }
133
134                 R.id.btn_division -> {
135                     result = a / b
136                 }
137             }
138
139             resultTv.text = "Result is $result"
140         }
141     }
```

activity_main.xml | MainActivity.kt | colors.xml

```
1   <?xml version="1.0" encoding="utf-8"?>
2   <resources>
3       <color name="black">#FF000000</color>
4       <color name="white">#FFFFFFFF</color>
5       <color name="orange">#FF9800</color>
6       <color name="gray">#878787</color>
7   </resources>
```

Screenshot/s of the Emulator:


Medium Phone API 36.1
1:30
Johns Calculator
Dark Mode
50
30
+
-
*
/
Result is 20.0
41%


1:30
Johns Calculator
Dark Mode
50
30
+
-
*
/
Result is 20.0
41%

Key Learnings:

<Provide your key learnings here – 3-5 bulleted sentences>